



MB 518: Decision analysis

Course Code	MB 518
Course Name	Decision analysis
L-T-P-C	2-0-0-2 (L-T-P-C)
Prerequisites	None
Intended for	MBA
Distribution	Compulsory
Approval	50th BoA (Jul 2023)

Course Contents

- **Module 1:** This module would set the context for decision analysis course. It would discuss a few illustrative examples in details (Eg Bidding problem, Pricing decision, Investment decision, outsourcing decision, Decision under uncertainty). (3 hours)
- **Module 2 Mathematical/Formal representation of consequences:** This module introduces a formal need and ways to measure the consequence of a decision alternate. In specific context explore the use of Linear functions, Piecewise Linear function, Loss functions, Quadratic functions and their relevance, roots of a Quadratic equation, Breakeven Price, exponential and logarithmic functions, Sequences (Geometric and Arithmetic) and functions of many variables. (4 hours)
- **Module 3 Review of Probability (Rapid):** Introduction to Probability and Random variables, Conditional probability, expected value, Summary measures, Fractiles, Measures of dispersion, Chebyshev's inequality, functions of random variables, Joint distribution of random variables, Covariance, Conditional expectations, Binominal, Poisson, and normal Distributions. (6 hours)
- **Module 4 Decision theory:** Method of sensitivity analysis, Method of breakeven analysis, Decision Problems under uncertainty, Decision trees, expected monetary value as a criterion, expected value of perfect information (EVPI), Structuring and solving sequential decision problem, case studies (2), sampling information, value of sample, optimal sample size to update prior probabilities. expected net gain in sampling, Case studies (2). Cash Equivalent, risk preference. (12 hours)
- **Module 5:** Loss functions and special structures (3 hours)

News boy problem and its variations

Lab Exercises (If applicable): Not applicable

Lab to be conducted on a 2-hour slot. It will be conducted in tandem with the theory course so the topics for problems given in the lab are already initiated in the theory class. The topics taught in the theory course should appropriately be sequenced for synchronization with the laboratory.

Text Books

1. VL Mote and T Madhavan, **Operations research**, Wiley Indian, 2016.
2. PG Moore and HM Thomas, **Anatomy of decisions**, Penguin Business, 1971